



Service Tool Instruction

Service Tool Instruction
Number: 3377541-5

Revision Level:

Released Date: 21-nov-
2003

Description

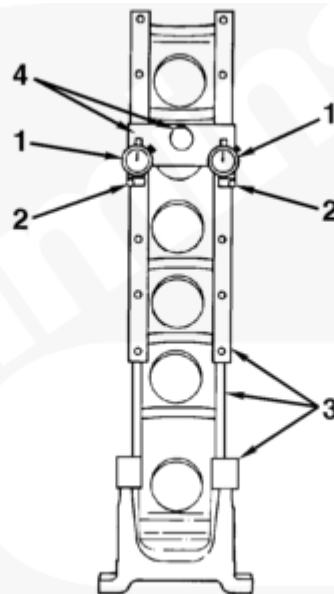
Connecting Rod Checking Fixture

Purpose

This document provides information for calibrating connecting rod checking fixture, Part Number ST-561. The fixture is used to check the length, bend, twist, and bore alignment of connecting rods. The fixture is used on the J & C, V-555, H, NH, NT, N14, V8-265, V-903, V6-140, V6-200, V8-185, V8-210, V8-350, VT8-430, V555, V-28, QST30, QSK45, QSK60, QSK23, QSK78, K19, K38, K50, L10, B, C, ISC, and ISL series engines.

The entire connecting rod checking fixture **must** be returned to the manufacturer if any wear points require service. Dial indicators and indicator contact points can be replaced in the field. Connecting rod checking guides are purchased separately.

©Cummins Inc



22600238

Table 1, Connecting Rod Checking Fixture, Part Number ST-561

Item Number	Part Number	Description	Quantity
1	ST-561-6	Dial indicator	2
2	3375093	Contact point	2
3	ST-561-5101	Base assembly	1

Table 1, Connecting Rod Checking Fixture, Part Number ST-561			
Item Number	Part Number	Description	Quantity
4	ST-561-5102	Indicator bracket assembly	1

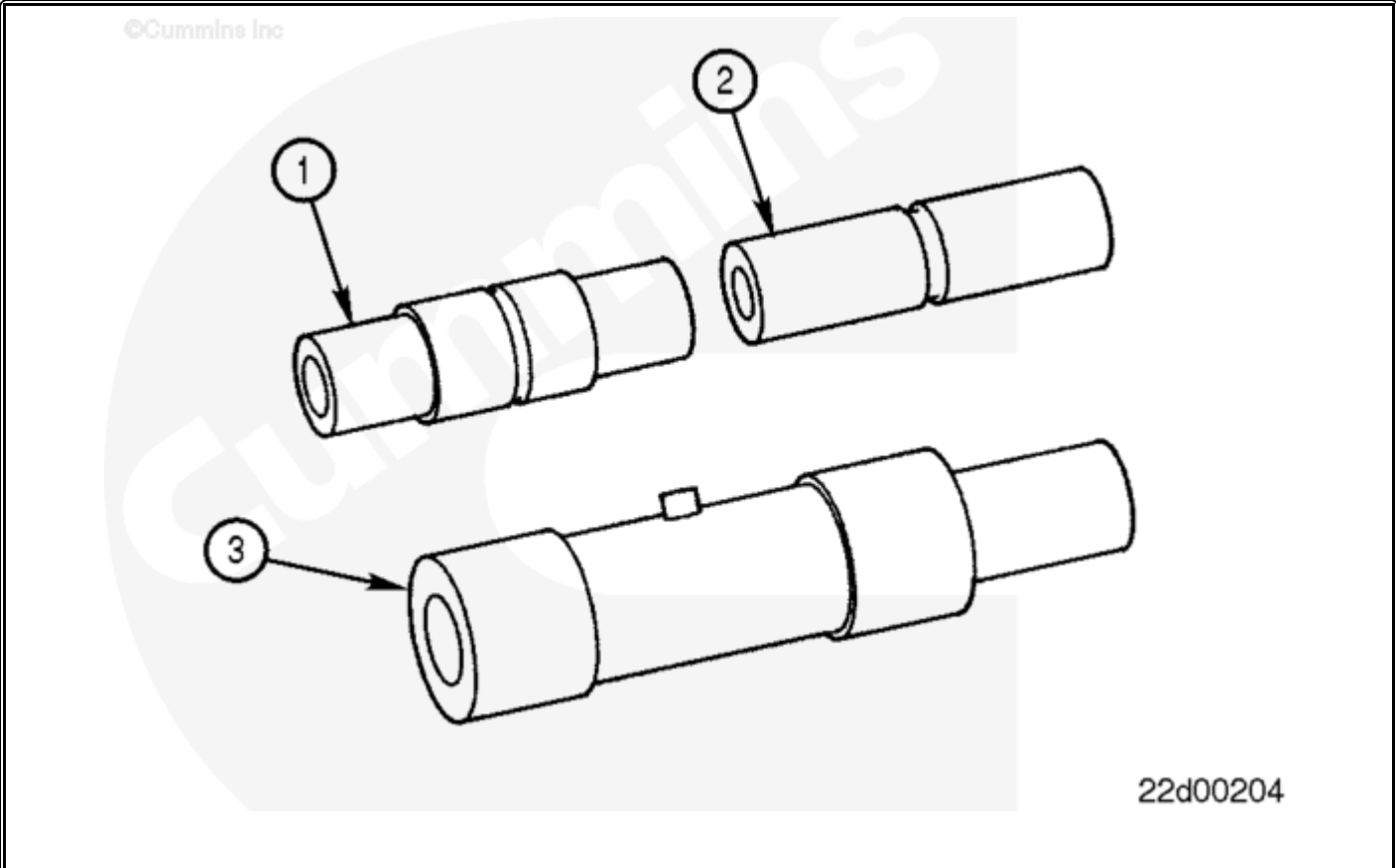


Table 2, Items Used with the Connecting Rod Checking Fixture, Purchased Separately			
Base Tool	Description of Connecting Rod Checking Guide	Engine Series	Item Number
ST-562	Complete kit	C and J	
ST-562-5001	Crankpin end	C and J	3
ST-562-2	Piston pin end without bushing	C, J, and V-555	1
ST-562-3	Piston pin end with bushing	C, J, and V-555	2
ST-562-9	Piston pin end without bushing	C	1

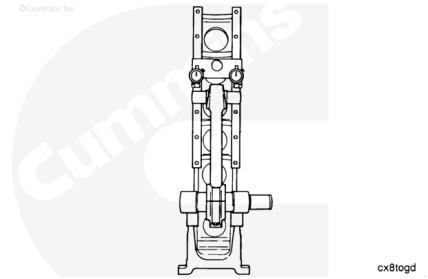
Table 2, Items Used with the Connecting Rod Checking Fixture, Purchased Separately			
Base Tool	Description of Connecting Rod Checking Guide	Engine Series	Item Number
Items			
ST-563	Complete kit (bolt and nut rods)	H, NH, and HT	
ST-563-5001	Crankpin end (bolt and nut rods)	H, NH, and HT	3
ST-563-2	Piston pin end without bushing	H, NH, HT, and V-28	1
ST-563-3	Piston pin end with bushing	H, NH, HT, V-12, and V-28	2
Items			
ST-758	Crankpin end (use with ST-563)	V-12 and V-28	3
Items			
3823287	Complete kit (bolt and nut rods)	NH and NT	
3823301	Crankpin end (bolt and nut rods)	NH and NT	3
ST-563-2	Piston pin end without bushing	NH and NT	1
ST-563-3	Piston pin end with bushing	NH and NT	2
Items			
3823785	Complete kit	N14	
3823786	Crank pin end	N14	3
3823787	Piston pin end without bushing	N14	1
3823788	Piston pin end with bushing	N14	2
Items			
ST-800	Complete kit	V6-200, V8-265, V8-350, VT8-430, and V903	
3375458	Crankpin end	V8-265 and V903	3

Table 2, Items Used with the Connecting Rod Checking Fixture, Purchased Separately			
Base Tool	Description of Connecting Rod Checking Guide	Engine Series	Item Number
ST-800-6	Piston pin end without bushing	V8-265 and V903	1
ST-800-7	Piston pin end with bushing	V8-265 and V903	2
Items			
ST-801	Complete kit	V6-140, V8-185, and V8-210	
ST-801-5001	Crankpin end	V6-140, V8-185, and V8-210	3
ST-801-2	Piston pin end without bushing	V6-140, V8-185, and V8-210	1
ST-801-3	Piston pin end with bushing	V6-140, V8-185, and V8-210	2
Items			
ST-1197	Crankpin end (use with ST-801)	V-555	3
Items			
3825185	Complete kit	QST30	
3825188	Crankpin end	QST30	3
3825187	Piston pin end without bushing	QST30	1
3825186	Piston pin end with bushing	QST30	2
Items			
3376690	Complete kit	L10, M11 and ISM	
3376691	Crankpin end	L10, M11 and ISM	3
3376692	Piston pin end without bushing	L10, M11 and ISM	1
3376693	Piston pin end with bushing	L10, M11 and ISM	2
Items			
3823285	Complete kit	B series	

Table 2, Items Used with the Connecting Rod Checking Fixture, Purchased Separately			
Base Tool	Description of Connecting Rod Checking Guide	Engine Series	Item Number
3823302	Crankpin end	B series	3
3823282	Piston pin end without bushing	B series	1
3823281	Piston pin end with bushing	B series	2
3823286	Complete kit	C series, ISC, and ISL	
3823303	Crankpin end	C series, ISC, and ISL	3
3823284	Piston pin end without bushing	C series, ISC, and ISL	1
3823283	Piston pin end with bushing	C series, ISC, and ISL	2
Items			
3165110	Complete kit	QSK45 and QSK60	
3163700	Crankpin end	QSK45 and QSK60	3
3163701	Piston pin end with bushing	QSK45 and QSK60	2
Items			
3165107	Crankpin end	QSK78	3
Items			
3165106	Piston pin end with bushing	QSK78 and QSK23	2
Items			
ST-1305	Complete kit	K19	
ST-1305-1	Crankpin end	K19	3
ST-1305-2	Piston pin end without bushing	K19, K38 and K50	1
ST-1305-3	Piston pin end with bushing	K19, K38 and K50	2
Items			
ST-1331	Crankpin end (use with ST-1305)	K38, K50, and QSK23	3

A connecting rod of known length, bend, and twist is called the reference connecting rod and is required to calibrate the fixture.

A new connecting rod for which the distance from the center of the crankshaft bore to the center of the piston pin bore (rod length) is known, can also be used.



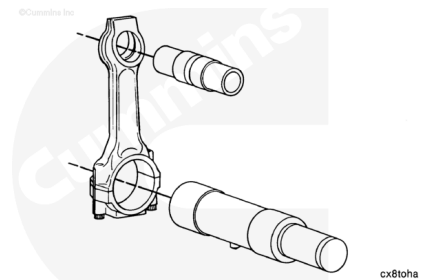
The connecting rod cap **must** be installed and torqued to the correct specifications or the calibration will be incorrect.

Install the appropriate connecting rod checking guide into the reference connecting rod piston pin bore and center the guide.

Install the appropriate connecting rod checking guide into the reference connecting rod crankshaft bore.

Make sure the pin on the guide is down and centered in the connecting rod.

Lock the guide in position by rotating the end of the guide.



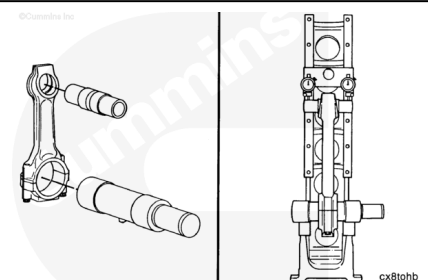
Install the reference connecting rod into the connecting rod checking fixture.

Loosen the knob and move the indicator bracket until both indicators touch the top of the piston pin guide.

Move the bracket toward the guide until the indicator needles have moved approximately 0.25 mm [0.010 in].

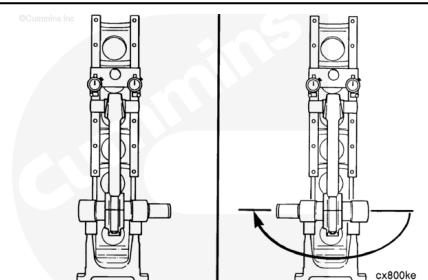
Turn the knob to tighten the bracket.

Adjust the indicator needles to zero. Move the connecting rod in and out to confirm the zero setting.



Remove the reference connecting rod from the fixture.

Rotate the connecting rod 180 degrees horizontally and install into the fixture.



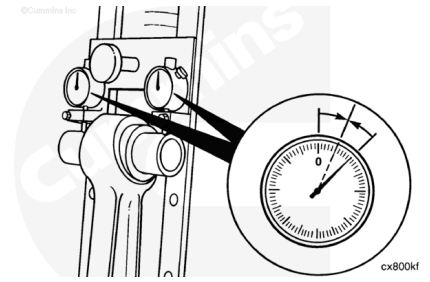
Be sure the indicators are at the zero setting.

If the indicator needles do **not** return to zero, adjust the indicator so the zero moves 1/2 of the indicated difference from the needle to the zero established during the previous step.

If the needle is more than 0.10 mm [0.004 in] from the zero established during the previous step, check for dirt or burrs on the connecting rod checking guides and fixture. If the zero is still **not** within the specification, check to make sure the reference connecting rod is **not** damaged.

Do **not** adjust the indicators after this step of the procedure or the measurements will be incorrect.

After completing the calibration of the fixture, remove the reference connecting rod.



Repeat the previous steps to install the connecting rod checking guides into the connecting rod to be measured.

Install the connecting rod into the connecting rod checking fixture.

Record the indicator reading. The difference of the calibrated zero of the indicators **must** be added or subtracted from the known length of the reference connecting rod to determine the length of the rod being measured.

Refer to the appropriate publication.

Procedure 001-014 in the Shop Manual C and J Engines, Bulletin 983627-DE.

Procedure 001-014 in the Shop Manual V/VT-378, V/VT-504, and V/VT-555 C.I.D. Engines, Bulletin 3379069.

Procedure 001-014 in the H and NH Series Shop Manual, Bulletin 3379067.

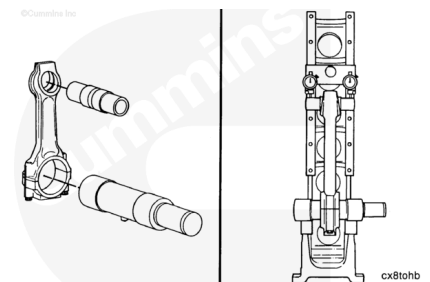
Procedure 001-014 in the Shop Manual V-28 Engines, Bulletin 3666014.

Procedure 001-014 in the N14 Engines 1991 Certification Levels, Bulletin 3810487.

Procedure 001-014 in the V/VT-903 and VTB-903 C. I. D. Engine Shop Manual, Bulletin 3379057.

Procedure 001-014 in the Shop Manual K38 and K50 Series Engine Volume II, Bulletin 3810304.

Procedure 001-014 in the Shop Manual L10 Series Engines 1991 Models, Bulletin 3810476.



Procedure 001-014 in the Shop Manual M11 Series, Bulletin 3666075.

Procedure 001-014 in the Shop Manual C Series Engines 1991 and 1994 Certification Levels, Bulletin 3666008.

Procedure 001-014 in the Shop Manual QSK45 and QSK60, Bulletin 4021373.

Procedure 001-014 in the Shop Manual QST30 Series Engine, Bulletin 3666190.

Last Modified: 12-Jan-2005
